SPLIT EXACT SEQUENCE FOR THE PICARD FUNCTOR WITH A SECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1027-1038 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1040-1093, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_pic.json (pb=2253, stmt 1027-1038, proof 1040-1093).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author notes '(small detail omitted)' in the descent application - named tiny check; the rescaling and f_ijk = 1 arguments are in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0B9L: sources/stacks-pic/pinned-original-source.tex, source lines 957–968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 023T: sources/stacks-pic/pinned-original-source.tex, source lines 1093–1093. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9M: sources/stacks-pic/pinned-original-source.tex, source lines 979–987. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be as in Definition 0B9L. Assume f has a section σ and that $\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism for all $T \in \mathrm{Ob}((Sch/S)_{fppf})$. Then*

$$0 \to \mathrm{Pic}(T) \to \mathrm{Pic}(X_T) \to \mathrm{Pic}_{X/S}(T) \to 0$$

is a split exact sequence with splitting given by $\sigma_T^* : \mathrm{Pic}(X_T) \to \mathrm{Pic}(T)$.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Denote $K(T) = \mathrm{Ker}(\sigma_T^* : \mathrm{Pic}(X_T) \to \mathrm{Pic}(T))$. Since σ is a section of f we see that $\mathrm{Pic}(X_T)$ is the direct sum of $\mathrm{Pic}(T)$ and $K(T)$. Thus by Lemma 0B9M we see that $K(T) \subset \mathrm{Pic}_{X/S}(T)$ for all T. Moreover, it is clear from the construction that $\mathrm{Pic}_{X/S}$ is the sheafification of the presheaf K. To finish the proof it suffices to show that K satisfies the sheaf condition for fppf coverings which we do in the next paragraph.

Let $\{T_i \to T\}$ be an fppf covering. Let $\mathcal{L}_i$ be elements of $K(T_i)$ which map to the same elements of $K(T_i \times_T T_j)$ for all i and j. Choose an isomorphism $\alpha_i : \mathcal{O}_{T_i} \to \sigma_{T_i}^* \mathcal{L}_i$ for all i. Choose an isomorphism

$$\varphi_{ij} : \mathcal{L}_i|_{X_{T_i \times_T T_j}} \longrightarrow \mathcal{L}_j|_{X_{T_i \times_T T_j}}$$

If the map

$$\alpha_j|_{T_i \times_T T_j} \circ \sigma_{T_i \times_T T_j}^* \varphi_{ij} \circ \alpha_i|_{T_i \times_T T_j} : \mathcal{O}_{T_i \times_T T_j} \to \mathcal{O}_{T_i \times_T T_j}$$

is not equal to multiplication by 1 but some u_{ij}, then we can scale φ_{ij} by u_{ij}^{-1} to correct this. Having done this, consider the self map

$$\varphi_{ki}|_{X_{T_i \times_T T_j \times_T T_k}} \circ \varphi_{jk}|_{X_{T_i \times_T T_j \times_T T_k}} \circ \varphi_{ij}|_{X_{T_i \times_T T_j \times_T T_k}} \quad \text{on} \quad \mathcal{L}_i|_{X_{T_i \times_T T_j \times_T T_k}}$$

which is given by multiplication by some regular function f_{ijk} on the scheme $X_{T_i \times_T T_j \times_T T_k}$. By our choice of φ_{ij} we see that the pullback of this map by σ is equal to multiplication by 1. By our assumption on functions on X, we see that $f_{ijk} = 1$. Thus we obtain a descent datum for the fppf covering $\{X_{T_i} \to X\}$. By Descent, Proposition 023T there is an invertible $\mathcal{O}_{X_T}$-module $\mathcal{L}$ and an isomorphism $\alpha : \mathcal{O}_T \to \sigma_T^* \mathcal{L}$ whose pullback to X_{T_i} recovers $(\mathcal{L}_i, \alpha_i)$ (small detail omitted). Thus $\mathcal{L}$ defines an object of $K(T)$ as desired. □